1 Solution — GIAM 3.3 Problem 1

1.1 Problem

Prove that if the cube of an integer is odd, then that integer is odd.

1.2 The Statement, Formalised

$$\forall x \in \mathbb{Z}, \quad x^3 \text{ is odd} \implies x \text{ is odd}.$$

1.3 Answer

$$\boxed{\text{Proved by contraposition: instead show } \forall x \in \mathbb{Z},\ x \text{ even} \implies x^3 \text{ even.}}$$

1.4 Proof

Theorem. *For every integer x, if x^3 is odd then x is odd.*

Proof. *This statement is logically equivalent to its contrapositive,*

$$\forall x \in \mathbb{Z}, \quad x \text{ is even} \implies x^3 \text{ is even},$$

so we prove that instead. (An integer is even or odd and never both, so "x is not odd" may be replaced by "x is even", and likewise for x^3.)

Suppose x is a particular but arbitrarily chosen integer, and suppose x is even. By the definition of even, there is an integer m with $x = 2m$. Cubing,

$$x^3 = (2m)^3 = 8m^3 = 2\left(4m^3\right).$$

> *Let $n = 4m^3$. Since m is an integer and the integers are closed under multiplication, n is an integer, and $x^3 = 2n$. So x^3 is even.*
>
> *Since x was arbitrary, x even $\implies x^3$ even holds for all integers x, and therefore so does the original statement.* ■

1.5 Why Not a Direct Proof?

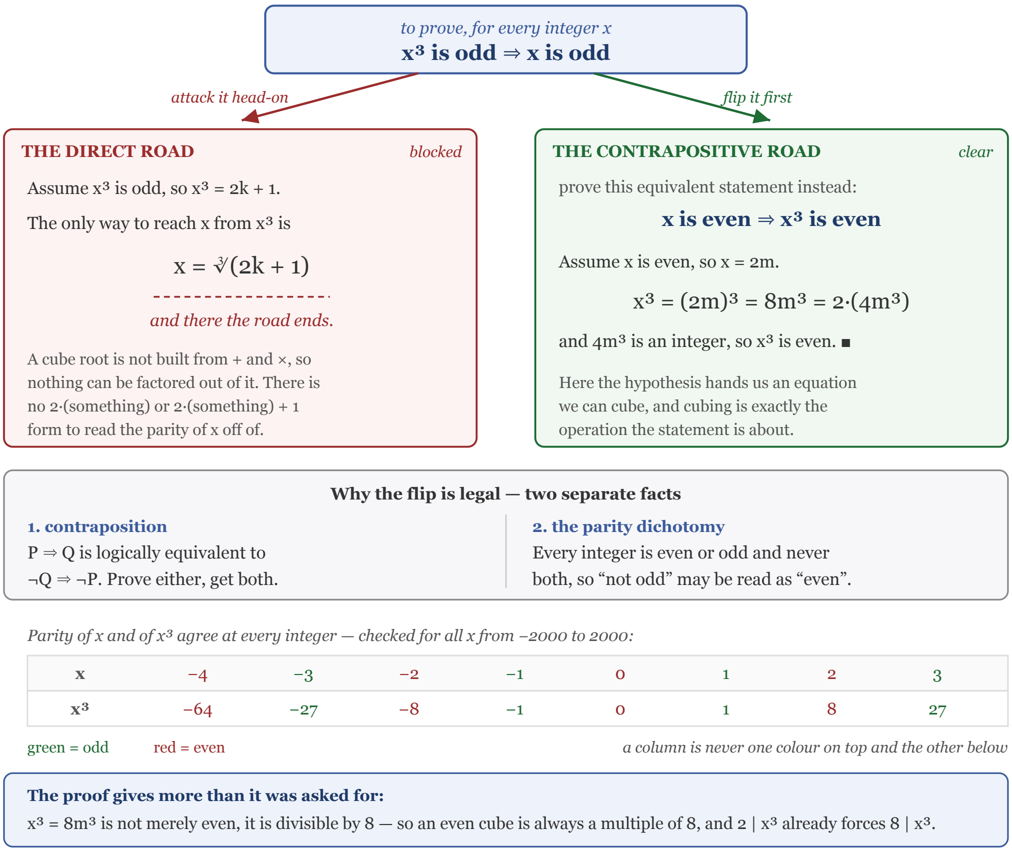

Figure 1.1: A diagram with the goal at the top, x cubed odd implies x odd, forking into two roads. The left road, in red and marked THE DIRECT ROAD – blocked, assumes x cubed is odd so x cubed equals 2k+1; the only way to reach x is x equals the cube root of 2k+1, and there the road ends, because a cube root is not built from plus and times, so nothing can be factored out of it and there is no 2 times something or 2 times something plus one form to read the parity off of. The right road, in green and marked THE CONTRAPOSITIVE ROAD – clear, proves the equivalent statement x even implies x cubed even: assume x is even so x equals 2m, then x cubed equals 8m cubed equals 2 times 4m cubed, and 4m cubed is an integer, so x cubed is even. It notes that here the hypothesis hands us an equation we can cube, and

cubing is exactly the operation the statement is about. A panel below gives the two separate facts making the flip legal: contraposition, that P implies Q is logically equivalent to not Q implies not P, and the parity dichotomy, that every integer is even or odd and never both, so ‘not odd’ may be read as ‘even’. A strip of sample values from minus four to three shows x and x cubed coloured by parity, with the columns never disagreeing, and notes the check ran over all x from minus 2000 to 2000. A final panel notes the proof gives more than asked: x cubed equals 8m cubed is divisible by 8, so an even cube is always a multiple of 8.

The direct attempt stalls immediately, and it is worth being precise about *where*. Assuming x^3 odd gives $x^3 = 2k+1$ for some integer k, and the only route back to x is $x = \sqrt[3]{2k+1}$. A cube root is not assembled from addition and multiplication, so there is no way to factor a 2 out of it — nothing in that expression can be massaged into the form $2n+1$.

The contrapositive reverses the flow of information. Now the hypothesis is about x itself, and it arrives as an *equation*, $x = 2m$. Equations can be cubed. And cubing is precisely the operation the theorem concerns, so one algebraic step lands directly on the conclusion.

1.6 Remark — The Contradiction Version

Every proof by contraposition can be recast as one by contradiction. The computational core, $x = 2m \Rightarrow x^3 = 2(4m^3)$, is *identical*; only the packaging differs:

	contraposition	**contradiction**
assumed	x even	x^3 odd *and* x even
derived	x^3 even	x^3 even
closes by	being the goal	clashing with the assumption

Written out, the right-hand column reads:

> ***Proof (by contradiction).*** *Suppose, to the contrary, that some integer x has x^3 odd but x even. Then $x = 2m$ for an integer m, so $x^3 = 8m^3 = 2\left(4m^3\right)$, which is even. But x^3 was assumed odd, and no integer is both. This contradiction shows no such x exists.* ■

Contraposition carries one assumption and ends on a positive statement; contradiction carries two and ends on a collision. When the contrapositive is available, it is usually the cleaner write-up — the contradiction version drags the unused hypothesis "x^3 is odd" along for the whole argument just to spring it at the end.

1.7 Remark — Where the Parity Dichotomy Is Doing Work

The contrapositive of "x^3 odd $\Rightarrow x$ odd" is literally

$$x \text{ is not odd} \implies x^3 \text{ is not odd},$$

and turning that into the statement actually proved requires a second, separate fact: **every integer is either even or odd, and never both.** That is what licenses reading "not odd" as "even".

This is easy to use without noticing, but it is a genuine theorem about $\mathbb{Z}$ (a consequence of the division algorithm: dividing by 2 leaves remainder 0 or 1, and exactly one of them). It also fails the moment the setting changes — over the rationals, $3/2$ is neither even nor odd, and the same "flip" would be invalid.

Note too that the dichotomy is used *twice*, once on x and once on x^3, and that "never both" is what closes the contradiction version.

1.8 Remark — The Theorem Is Really an Equivalence

The converse, x odd $\Rightarrow x^3$ odd, is also true and equally quick: $x = 2k+1$ gives

$$x^3 = 8k^3 + 12k^2 + 6k + 1 = 2\left(4k^3 + 6k^2 + 3k\right) + 1.$$

So in fact

$$\forall x \in \mathbb{Z}, \quad x^3 \text{ is odd} \iff x \text{ is odd},$$

and by the same argument x^n is odd $\iff$ x is odd for every $n \geq 1$ — verified exhaustively for $1 \leq n \leq 12$ over $-200 \leq x \leq 200$. There is nothing special about the exponent 3; the cube is just the smallest case where the direct approach is visibly hopeless.

The clean way to see the general statement: reduction mod 2 respects both addition and multiplication, so it sends x^n to $\bar{x}^{\,n}$ in $\mathbb{Z}/2\mathbb{Z}$. In that two-element system $\bar{0}^{\,n} = \bar{0}$ and $\bar{1}^{\,n} = \bar{1}$ — powers never change parity, for any exponent. The whole family of statements collapses to that one observation.

1.9 Remark — What the Proof Proves Without Being Asked

The contrapositive proof produces $x^3 = 8m^3$, which is stronger than “even”. An even cube is always a multiple of 8, so for integers

$$2 \mid x^3 \implies 8 \mid x^3.$$

This is the same kind of free strengthening that showed up in 3.1 Problem 7, where the witness turned out to be a product of consecutive integers. Writing out the algebra rather than stopping at the first sufficient conclusion tends to hand you these.

There is a contrast worth drawing with squares, though. Odd *squares* are heavily constrained: every one is $\equiv 1 \pmod{8}$. Odd *cubes* are not constrained at all — cubing permutes the odd residues mod 8, and indeed mod 2^k for every k (checked through 2^6), so odd cubes realise all four odd classes $1, 3, 5, 7$. The strengthening here lives entirely on the even side.